

# Basic Electrical Engineering (BT EE103A )



हरियाणा केंद्रीय विश्वविद्यालय  
**CENTRAL UNIVERSITY OF HARYANA**  
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-III

Part-II

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## CONSTRUCTIONAL FEATURES OF A THREE-PHASE INDUCTION MOTOR

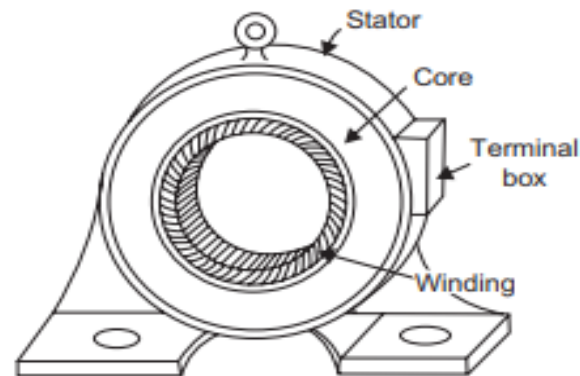
A three-phase induction motor consists of two main parts, namely stator and rotor.

1. **Stator:** It is the stationary part of the motor. It has three main parts, namely outer frame, stator core, and stator winding.

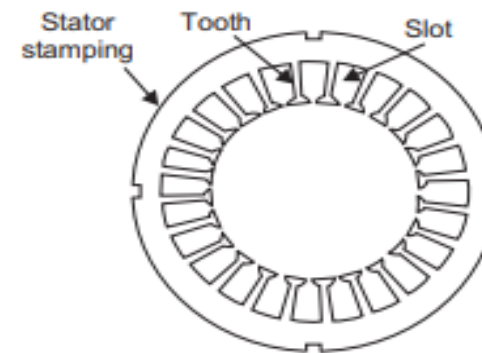
- (a) **Outer frame:** It is the outer body of the motor. Its function is to support the stator core and to protect the inner parts of the machine. For small machines, the frame is casted, but for large machines, it is fabricated.

To place the motor on the foundation, feet are provided in the outer frame as shown in Figure 12.1.

- (b) **Stator core:** The stator core is to carry the alternating magnetic field which produces hysteresis and eddy current losses; therefore, core is built up of high grade silicon steel stamping. The stampings are assembled under hydraulic pressure and are keyed to the frame. Each stamping is insulated from the other with a thin varnish layer. The thickness to the stamping usually varies from 0.3 to 0.5 mm. Slots are punched on the inner periphery of the stampings, as shown in Figure 12.2, to accommodate stator winding.



**Fig. 12.1** Stator of 3-phase induction motor

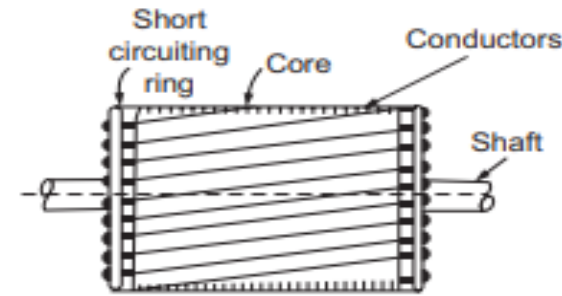


**Fig. 12.2** Stator stamping

- (c) **Stator winding:** The stator core carries a three-phase winding which is usually supplied from a three-phase supply system. The six terminals of the winding (two of each phase) are connected in the terminal box of the machine. The stator of the motor is wound for definite number of poles, the exact number being determined by the requirement of speed. It will be observed that greater the number of poles, the lower is the speed and vice-versa, since  $N_s \propto \frac{1}{P} \left( \text{or } N_s = \frac{120f}{P} \right)$ . The three-phase winding may be connected in star or delta externally through a starter.

2. **Rotor:** It is the rotating part of the motor. There are two types of rotors, which are employed in three-phase induction motors, namely squirrel-cage rotor and phase-wound rotor.

- (a) **Squirrel-cage rotor:** The motors employing this type of rotor are known as 'squirrel-cage induction motors'. Most of the induction motors are of this type because of simple and rugged construction of rotor. A squirrel-cage rotor consists of a laminated cylindrical core having semi-closed circular slots at the outer periphery. Copper or aluminium bar conductors are placed in these slots and short circuited at each end by copper or aluminium rings, called short-circuiting rings, as shown in Figure 12.3. Thus, the rotor winding is permanently short circuited, and it is not possible to add any external resistance in the rotor circuit.

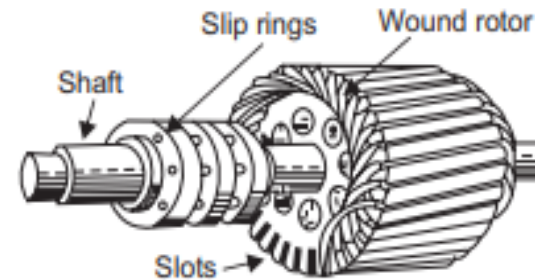


**Fig. 12.3 Squirrel cage rotor**

The rotor slots are usually not parallel to the shaft but are skewed. Skewing of rotor has the following advantages:

- (i) It reduces humming, thus ensuring quiet running of a motor.
- (ii) It results in a smoother torque curves for different positions of the rotor.
- (iii) It reduces the magnetic locking of the stator and rotor.
- (iv) It increases the rotor resistance due to the increased length of the rotor bar conductors.

- (b) **Phase-wound rotor:** Phase-wound rotor is also called slip-ring rotor and the motors employing this type of rotor are known as 'phase-wound or slip-ring induction motors'. Slip-ring rotor consists of a laminated cylindrical core having semi-closed slots at the outer periphery and carries a three-phase insulated winding. The rotor is wound for the same number of poles as that of stator. The three finish terminals are connected together forming star point, and the three start terminals are connected to three copper slip-rings fixed on the shaft (Figure 12.4).

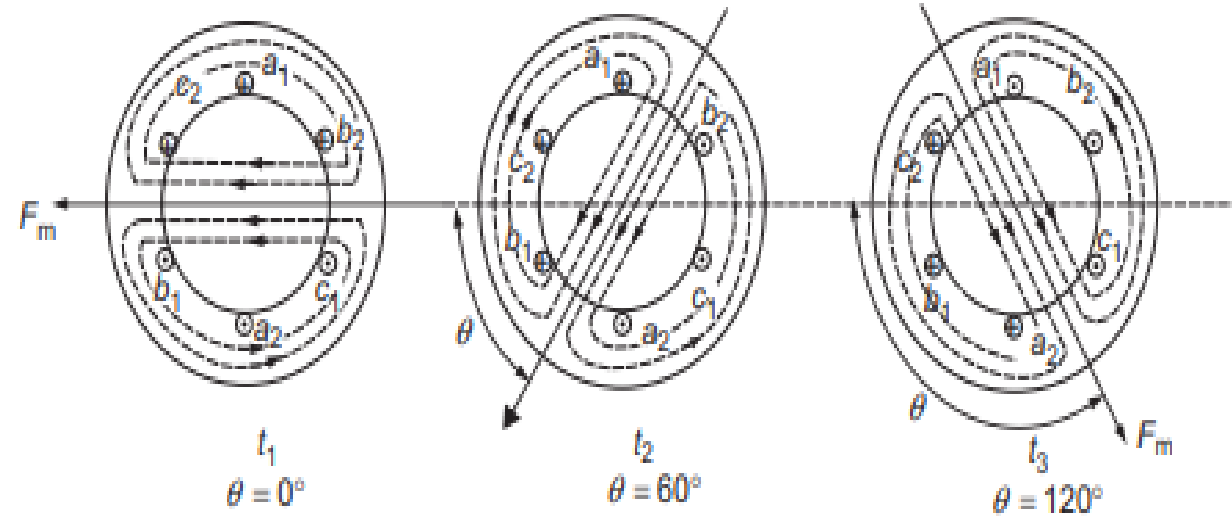
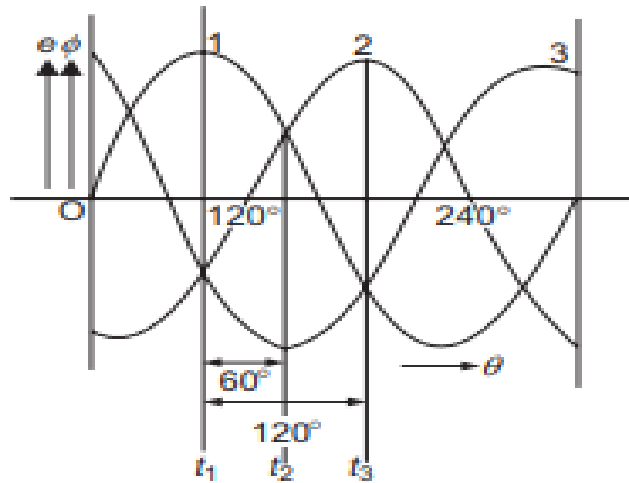


**Fig. 12.4 Phase-wound rotor**

In this case, depending upon the requirement, any external resistance can be added in the rotor circuit. In this case also, the rotor is skewed.

A mild steel shaft is passed through the centre of the rotor and is fixed to it with key. The purpose of shaft is to transfer mechanical power.

## PRODUCTION OF REVOLVING FIELD

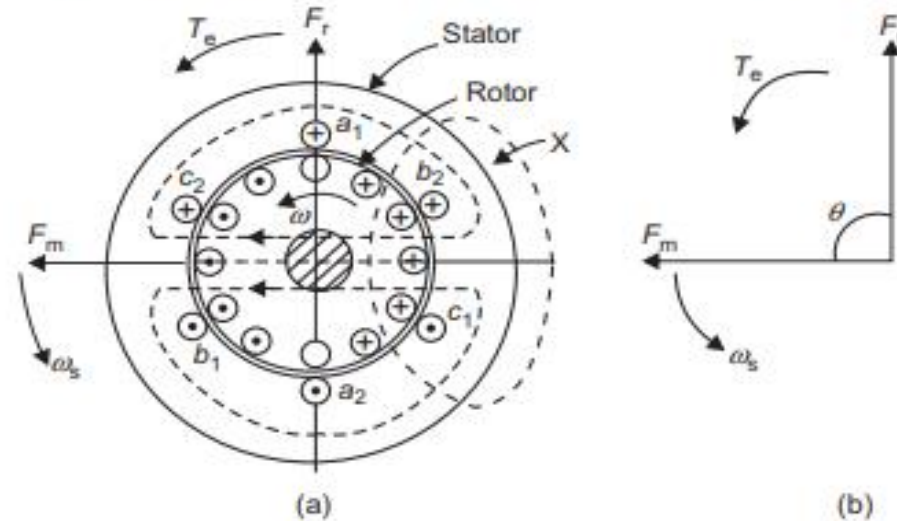


In this case, we have observed that when supply from phase 1, 2, and 3 is given to coil  $a_1a_2, b_1b_2$ , and  $c_1c_2$ , respectively, an anticlockwise rotating field is produced. If the supply to coil  $a_1a_2, b_1b_2$ , and  $c_1c_2$  is given from phase 1, 3, and 2, respectively, the direction of rotating field is reversed. Therefore, to reverse the direction of rotation of rotating field, the connections of any two supply terminals are inter changed.



## PRINCIPLE OF OPERATION

The direction of resultant field is marked by an arrow head  $F_m$ . Let this field is rotating in an anticlockwise direction at an angular speed of  $\omega_s$  radians per second, that is, synchronous speed.



**Fig. 12.9** (a) Induced emf/current in rotor conductors at an instant  
(b) Phasor representation of stator and rotor field at an instant

The stationary rotor conductors cut the revolving field and due to electromagnetic induction an emf is induced in the rotor conductors. As the rotor conductors are short circuited, current flows through them in the direction as marked in the figure. Rotor current carrying conductors set up a resultant field  $F_r$ . This field tries to come in line with the stator main field  $F_m$ . Due to this, an electromagnetic torque  $T_e$  is developed in the anticlockwise direction. Therefore, rotor starts rotating in same direction in which stator field is revolving.

## SLIP

The rotor of an induction motor always rotates at a speed less than synchronous speed. The difference between the flux speed ( $N_s$ ) and the rotor speed ( $N$ ) is called slip. It is usually expressed as a percentage of synchronous speed ( $N_s$ ) and is represented by symbol  $S$ .

Mathematically,

$$\% \text{ slip,} \quad \% S = \frac{N_s - N}{N_s} \times 100$$

$$\text{or fractional slip,} \quad S = \frac{N_s - N}{N_s}$$

$$\text{Rotor speed,} \quad N = N_s (1 - S)$$

The difference between synchronous speed and rotor speed is called slip speed, that is,

$$\text{Slip speed} = N_s - N$$

The value of slip at full-load varies from about 6 per cent small motors to about 2 per cent for large motors.

## Importance of Slip

Slip plays an important role in the operation of an induction motor. We have already seen that the difference between the rotor speed and synchronous speed of flux determine the rate at which the flux is cut by rotor conductors and hence the magnitude of induced emf, that is,  $e_2 \propto N_s - N$

Rotor current,  $i_2 \propto e_2$  and torque,  $T \propto i_2$

$$T = K(N_s - N)$$

or 
$$T = KN_s \left( \frac{N_s - N}{N_s} \right)$$

or 
$$T = K_1 S$$

Hence 
$$T \propto S$$

Thus, greater the slip greater will be the induced emf or rotor current, and hence, larger will be the torque developed.

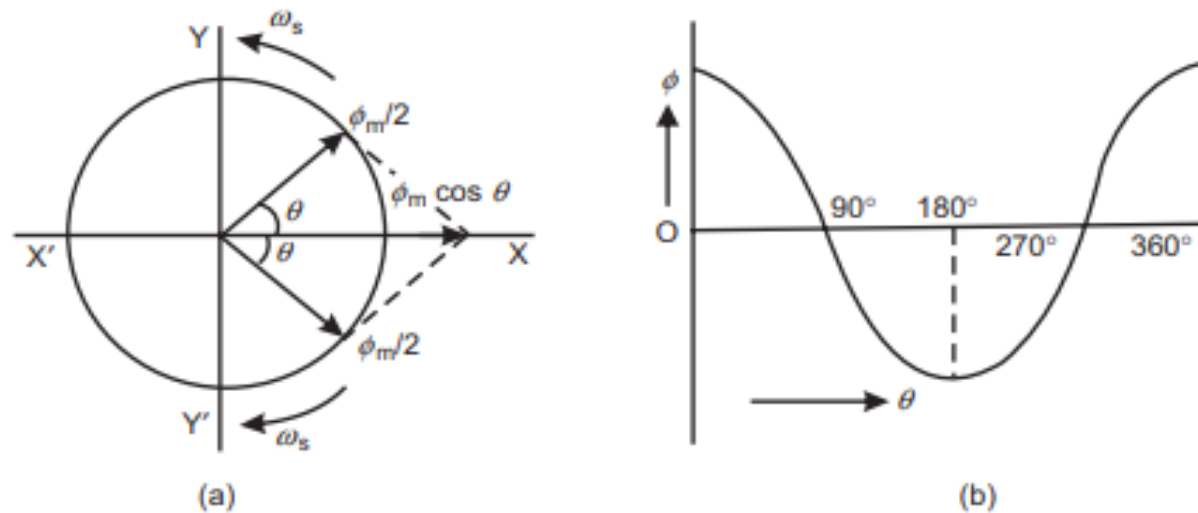
At no-load, induction motor requires small torque to meet with the mechanical, iron, and other losses, and therefore, slip is small. When the motor is loaded, greater torque is required to drive the load, and therefore, the slip increases and rotor speed decreases slightly.

Therefore, it is observed that slip in an induction motor adjusts itself to such a value to meet the required driving torque under normal operation.



## NATURE OF FIELD PRODUCED IN SINGLE-PHASE INDUCTION MOTORS

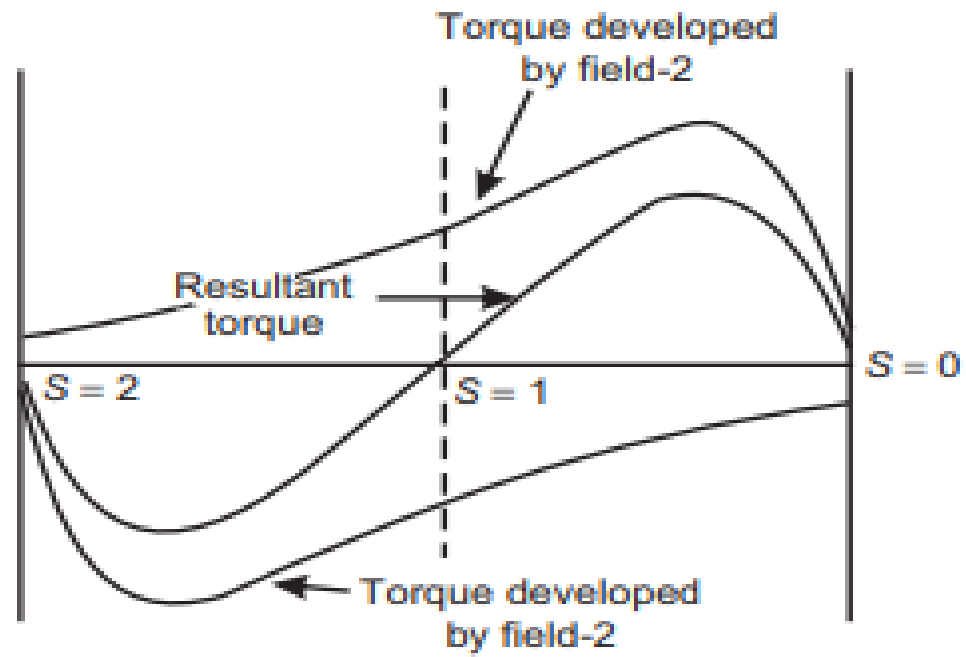
Let  $\phi_m$  be the pulsating field that has two components each of magnitude  $\phi_m/2$ . Both are rotating at the same angular speed  $\omega_s$  rad/s but in opposite direction as shown in Figure 13.1(a). The resultant of the two fields is  $\phi_m \cos \theta$ . This shows that resultant field varies according to cosine of the angle  $\theta$ . The wave shape of the resultant field is shown in Figure 13.1(b).



**Fig. 13.1** (a) Two vectors each of magnitude  $\phi_m/2$  rotating in opposite direction at same angular velocity (b) Wave shape of resultant field

Therefore, an alternating field can be represented by the two fields each of half the magnitude rotating at same angular speed of  $\omega_s$  rad/s but in opposite direction.

## TORQUE PRODUCED BY SINGLE-PHASE INDUCTION MOTOR



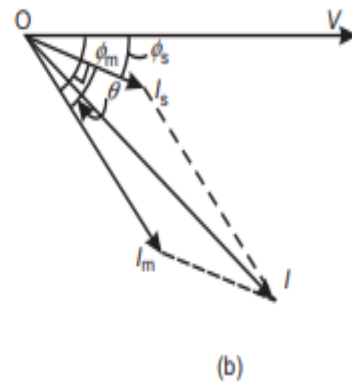
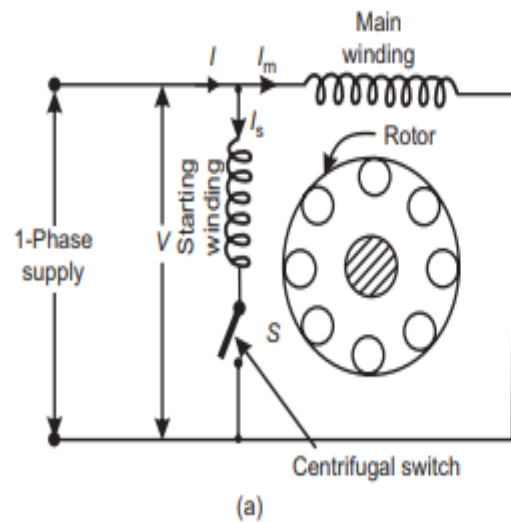
## TYPES OF MOTORS

Various methods (means) are employed to obtain the starting torque in single-phase induction motors. Accordingly, they are classified as follows:

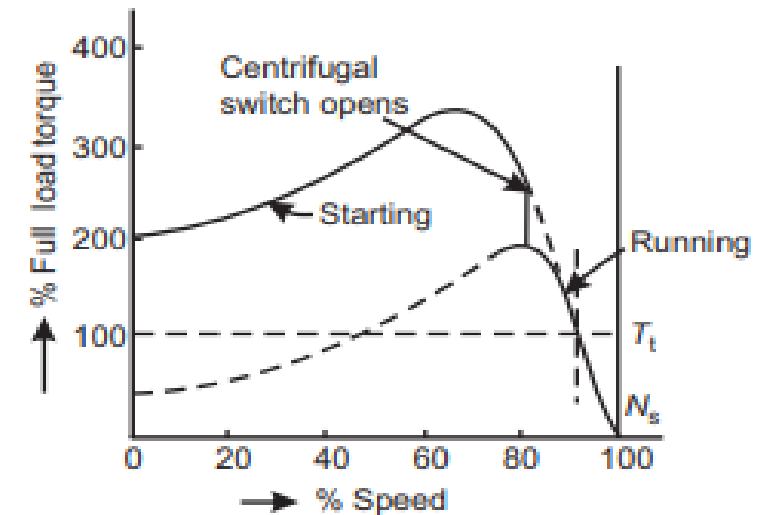
1. Split-phase motors
2. Shaded pole motors
3. AC series motors or Commutator motors.

### 1. Split-phase motors

In split phase motors, the single-phase winding is split into two parts, one of them is called a starting winding and the other is called a running winding.



$$N_s = \frac{120f}{P} \text{ rpm}$$

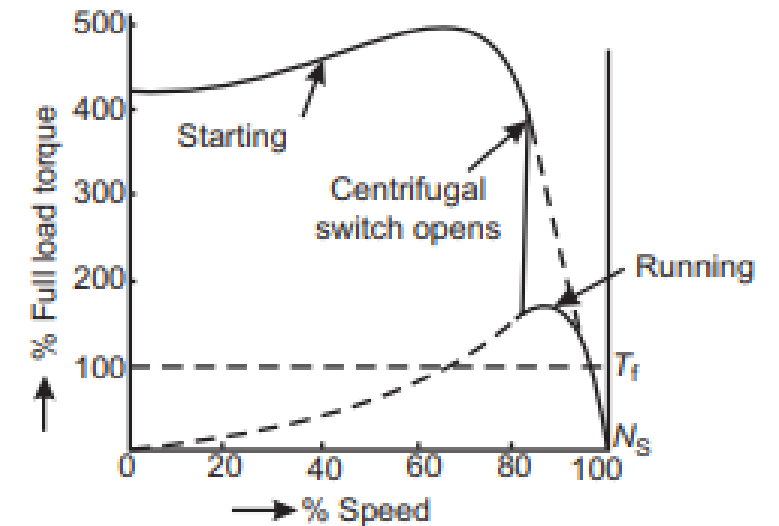
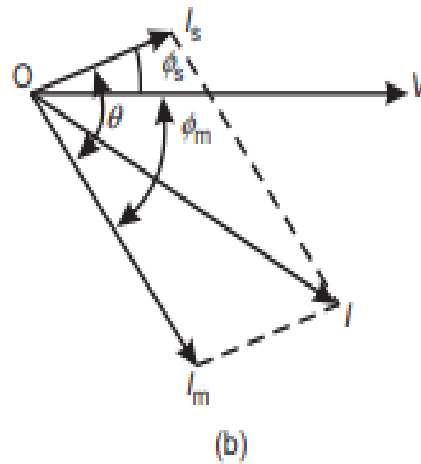
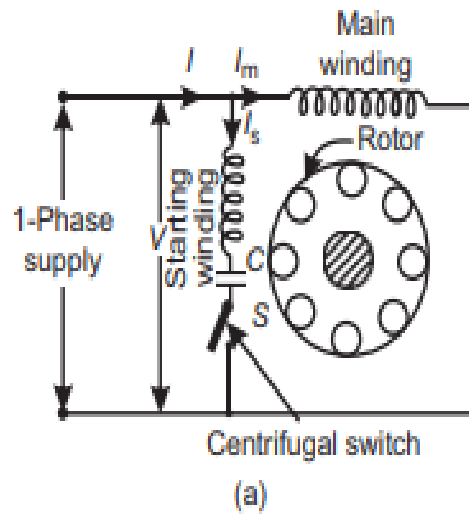


## CAPACITOR MOTORS

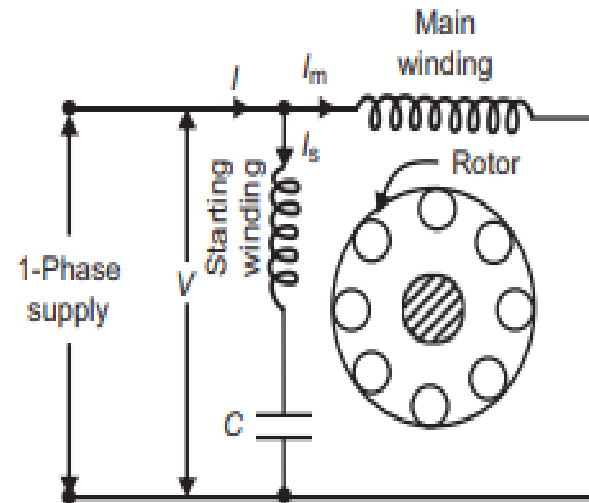
It is also a split-phase motor. Accordingly, capacitor motors may be classified as follows:

1. Capacitor start motors
2. Capacitor run motors
3. Capacitor start and capacitor run motors.

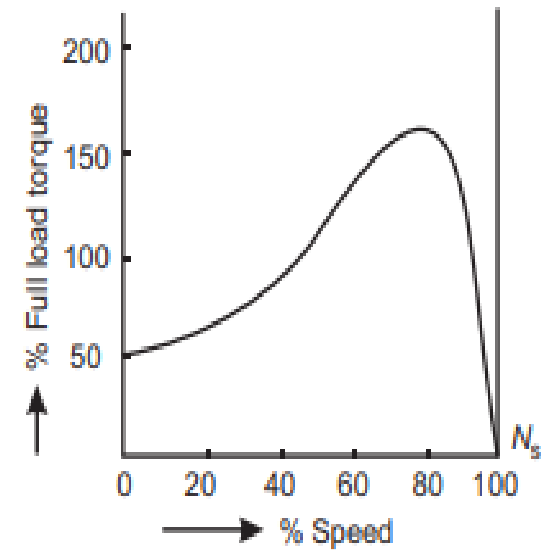
### 1. Capacitor start motors



## 2. Capacitor run motors

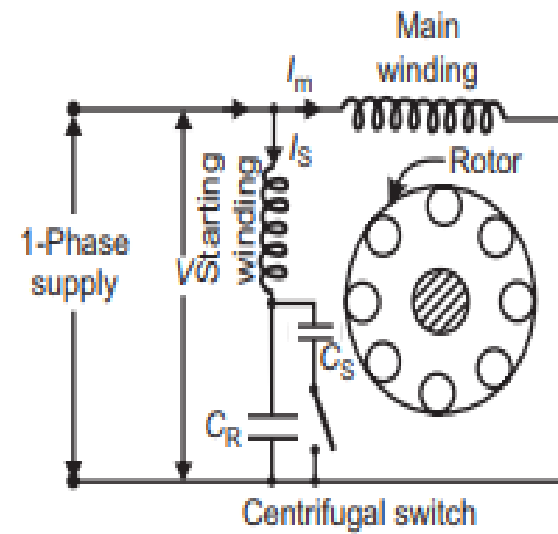


(a)

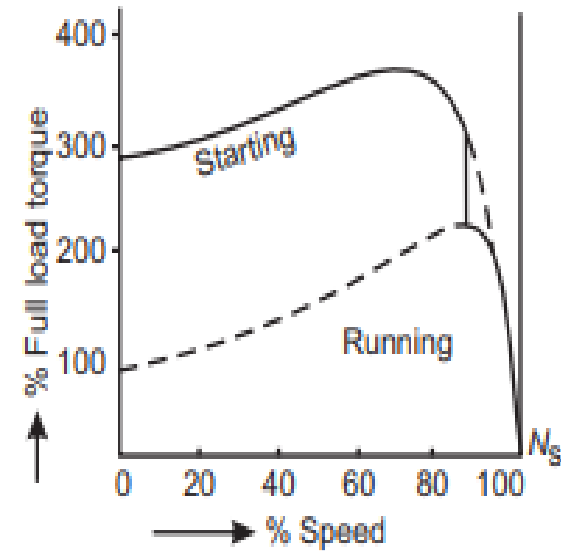


(b)

### 3. Capacitor start and capacitor run motors.



(a)



(b)





THANKS